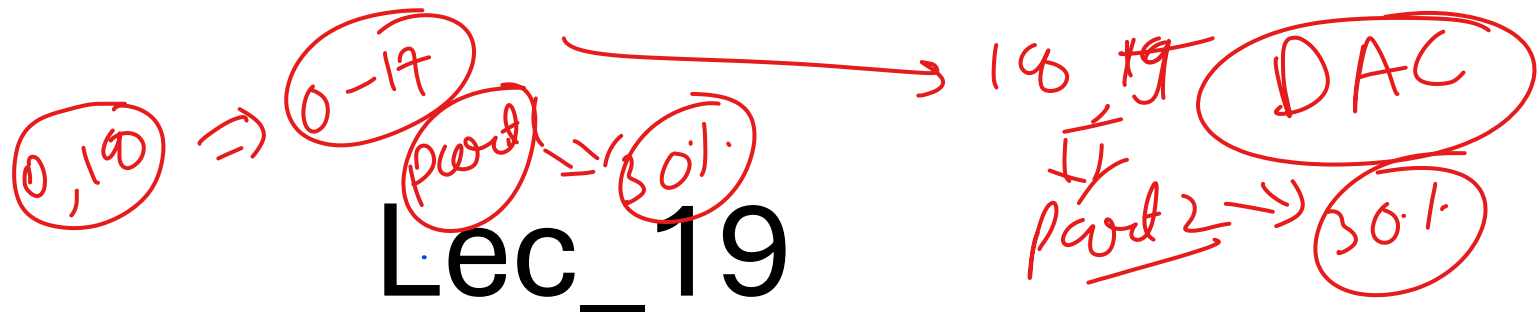
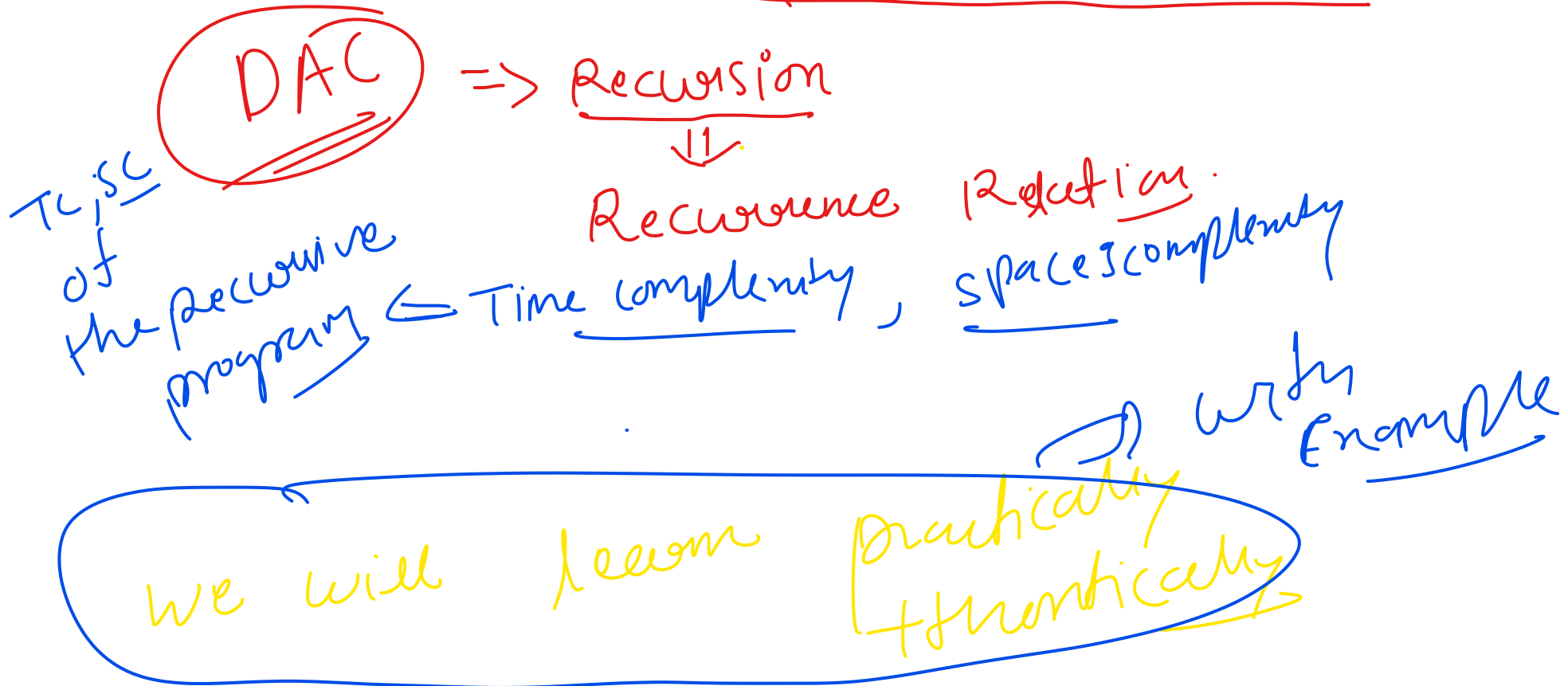




Lec_19

Introduction to Recurrence Relation



Write Recurrence relation

```
1.) fact( int n)
{ if(n<=1)
  return 1;
else
  return(n*fact(n-1))
}
```

$\Rightarrow$ $n \times \text{fact}(n-1)$

$\Rightarrow$ recursion

fact(n)

fact(n-1)

Let's Time complexity of fact(n)
is $T(n)$

$$\underline{T(n)} = C + T(n-1)$$

$$\underline{T(n) = T(n-1) + C}$$

RA

Recurrence Relation

If $\text{fact}(n) = T(n)$
then $\text{fact}(n-1) = T(n-1)$

Write Recurrence relation

2.) A(int n)

```
{ if(n<=1)
  return (n^2+n+10);
else
  return 10* A(n/2)*A(n/2)*A(n/2);
}
```

fix
constant

3 times

10 * A(n/2) * A(n/2) * A(n/2)

10 times recur
will be call

$T(n) = 3T(n/2) + C$
12A

Ans TC of A(n) is $T(n)$

$$T(n) = C + 3T(n/2)$$

if A(n) is $T(n)$
then $A(n/2) \Rightarrow T(n/2)$

Let for $A(n)$, TC is $T(n)$

$$T(n) = C + A(n/2)$$

Write Recurrence relation

3.) $A(\text{int } n)$

```
{ if(n<=1)
  return (n^2+n+10);
else
  return 10*A(n/2)+n^2+n^3;
}
```

small

$$T(n) = T(n/2) + C$$

$$T(n) = T(n/2) + C$$

Write Recurrence relation

```
4.) A(int n)
{ if(n <= 25)
  return n;
else
{
  for(int i = 1; i <= 4; i++)
    A(n/2);
  for(int i = 1; i <= n^2; i++)
    x = y + z;
}
```

Let, $T(n)$ is T_c for $A(n)$

$$T(n) = c + 4T(n/2) + n^2$$

n^2
canceled c

$A(n) \Rightarrow T(n)$
 $A(n/2) \Rightarrow T(n/2)$

$$T(n) = 4T(n/2) + n^2 + c$$

$n^2 > c$

$$T(n) = 4T(n/2) + n^2$$

$T(n) = TC$ for the $A(n)$

5.)

Write Recurrence relation

A(int n)

{ return n^n ;

}

10^{10}

88

$T(n) = C$

Constant

Thanks

Join telegram
channel

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